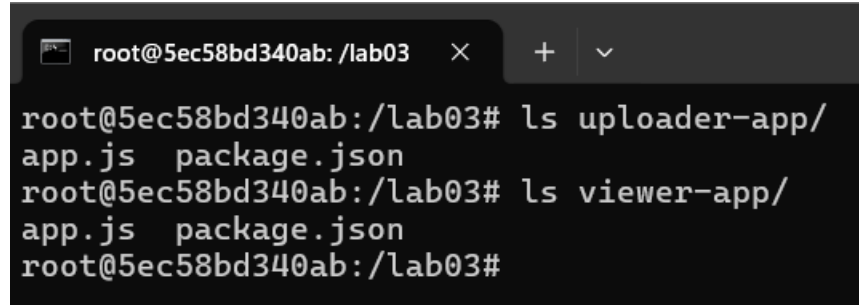
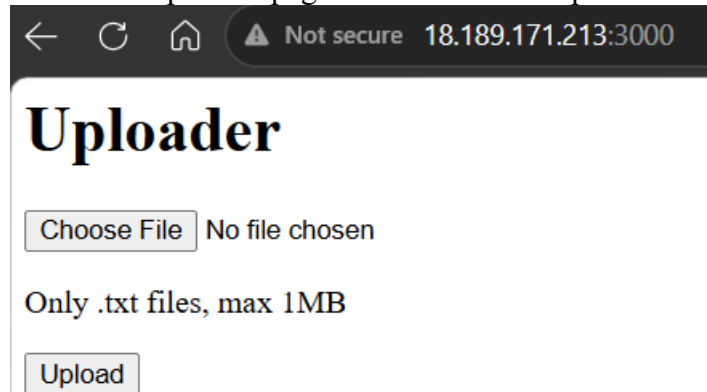


1. Screenshot of your local lab03/uploader-app/ and lab03/viewer-app/ folders showing in each.

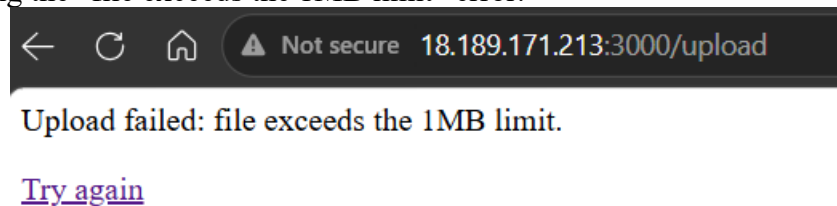


```
root@5ec58bd340ab: /lab03  ×  +  ▾
root@5ec58bd340ab:/lab03# ls uploader-app/
app.js  package.json
root@5ec58bd340ab:/lab03# ls viewer-app/
app.js  package.json
root@5ec58bd340ab:/lab03#
```

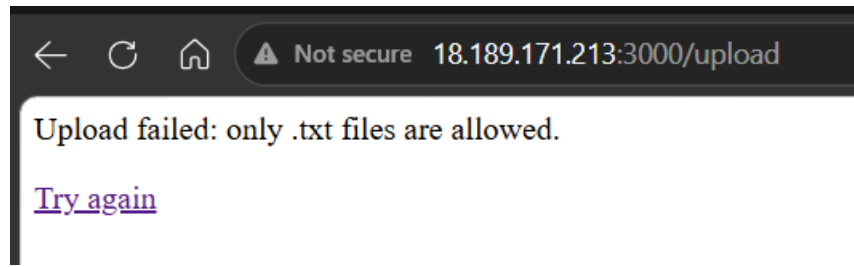
- a.
2. Screenshot of the Uploader page and a successful upload confirmation.



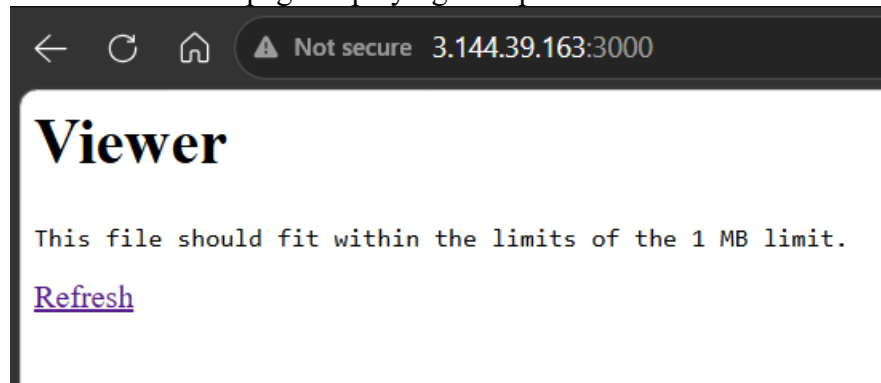
- a.
- b.
3. Screenshot of attempting to upload a app.js and package.json .txt file larger than 1MB, showing the “file exceeds the 1MB limit” error.



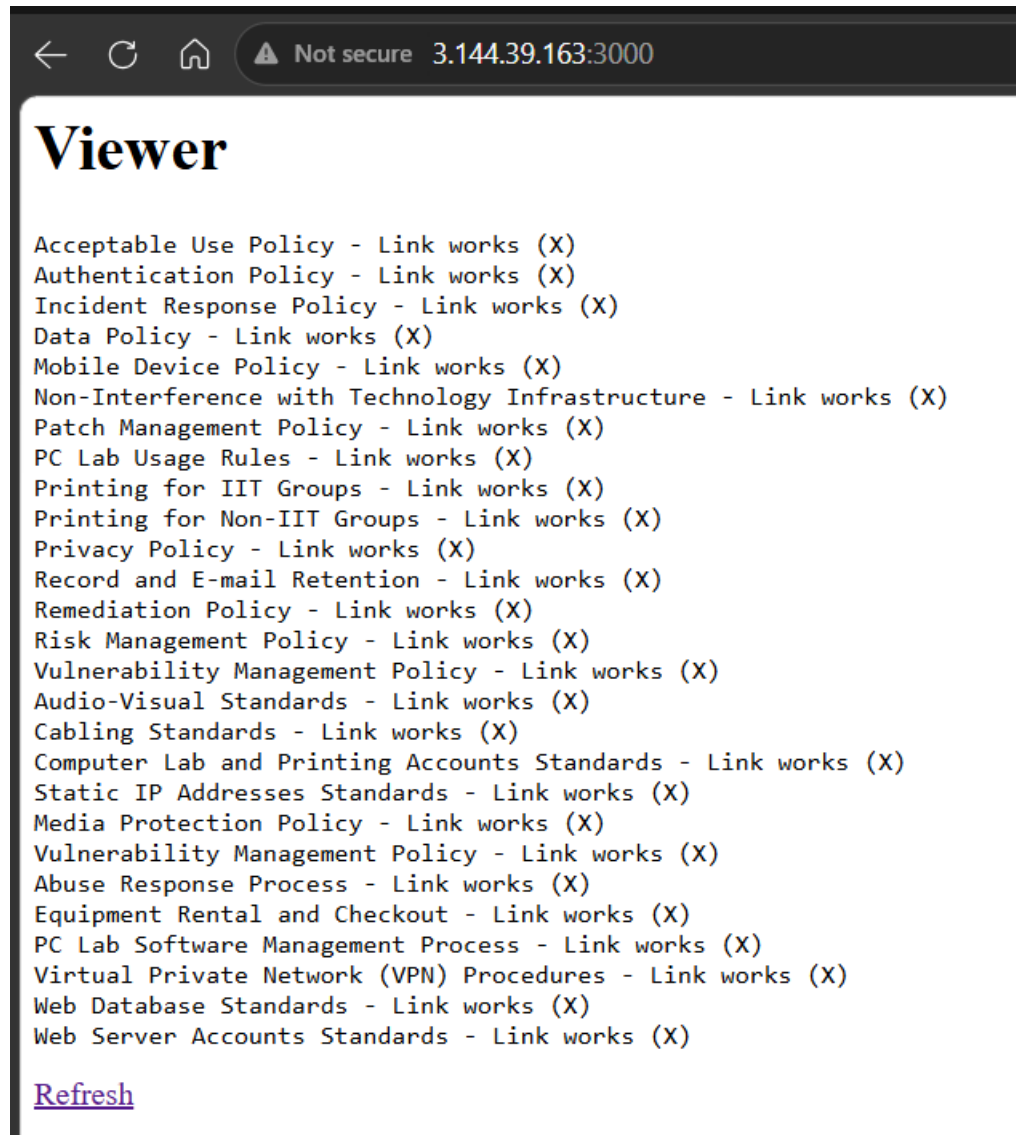
- a.
4. Screenshot of attempting to upload a non-.txt file (e.g., a .jpg or .pdf), showing the “only .txt files are allowed” error.



- a.
5. Screenshot of the Viewer page displaying the uploaded file's content.



- a.
6. Screenshot of the Viewer page after uploading a second, different file and clicking Refresh, showing the updated content.



a.

7. Screenshot of aws ec2 describe-launch-templates showing both templates.

```
root@5ec58bd340ab:/# aws ec2 describe-launch-templates --region us-east-2 \
--query "LaunchTemplates[*].[LaunchTemplateName,LatestVersionNumber]" \
--output table
```

DescribeLaunchTemplates	
itmo-444-544-viewer-lt	1
itmo-444-544-uploader-lt	1

a.

```
root@5ec58bd340ab:/#
```

8. Screenshot of aws ec2 describe-launch-template-versions --launch-template-name <uploader-lt-name> after re-running the create script twice, showing more than one version.

```
root@5ec58bd340ab:/# aws ec2 describe-launch-template-versions --launch-template-name itmo-444-544-uploader-lt --region us-east-2 --query "LaunchTemplateVersions[*].[VersionNumber,CreateTime]" --output table
```

DescribeLaunchTemplateVersions	
2	2026-09-30T21:35:58+00:00
1	2026-09-30T21:32:12+00:00

a.

```
root@5ec58bd340ab:/#
```

9. Screenshot of aws iam get-role-policy output (or screenshot) for both roles, showing each has only its one intended action on the exact shared.txt key.

```

aws iam get-role-policy \
  --role-name itmo-444-544-viewer-role \
  --policy-name itmo-444-544-viewer-role-policy
{
  "RoleName": "itmo-444-544-uploader-role",
  "PolicyName": "itmo-444-544-uploader-role-policy",
  "PolicyDocument": {
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Action": [
          "s3:PutObject"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-hvtran21-filesshare/shared.txt"
      }
    ]
  }
}
{
  "RoleName": "itmo-444-544-viewer-role",
  "PolicyName": "itmo-444-544-viewer-role-policy",
  "PolicyDocument": {
    "Version": "2012-10-17",
    "Statement": [
      {
        "Effect": "Allow",
        "Action": [
          "s3:GetObject"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-hvtran21-filesshare/shared.txt"
      },
      {
        "Effect": "Allow",
        "Action": [
          "s3:ListBucket"
        ],
        "Resource": "arn:aws:s3:::itmo-444-544-hvtran21-filesshare"
      }
    ]
  }
}
root@5ec58bd340ab:/#

```

a.

10. Screenshot of create_app_stack.sh output showing both instances running with public IPs.

```

Waiting for both instances to reach 'running'...
-----
| DescribeInstances |
+-----+-----+
| viewer-app | 3.144.39.163 |
| uploader-app | 18.189.171.213 |
+-----+-----+

```

a.

Instances are running. Give them ~60-90s to finish installing Node.js before browsing to :3000.

11. Screenshot of delete_app_stack.sh completing with no errors.

```

==      Deleting      launch   templates      ==
{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-08d0ebc15ec6eda61",
    "LaunchTemplateName": "itmo-444-544-uploader-lt",
    "CreateTime": "2026-09-30T21:18:13+00:00",
    "CreatedBy": "arn:aws:iam::417873488887:user/henry-tran-lab-user",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      "Managed": false
    }
  }
}

{
  "LaunchTemplate": {
    "LaunchTemplateId": "lt-0e25847acb8ccdd2c",
    "LaunchTemplateName": "itmo-444-544-viewer-lt",
    "CreateTime": "2026-09-30T21:18:15+00:00",
    "CreatedBy": "arn:aws:iam::417873488887:user/henry-tran-lab-user",
    "DefaultVersionNumber": 1,
    "LatestVersionNumber": 1,
    "Operator": {
      "Managed": false
    }
  }
}

==      Removing      IAM      instance      profiles      and      roles      ==
==      Emptying      and      deleting      S3      bucket      ==
delete: s3://itmo-444-544-hvtran21-fileshare/shared.txt
Cleanup complete.

```

- a. Cleanup complete.
12. One paragraph explaining why the uploader and viewer use separate IAM roles instead of one shared role with both permissions.
 - a. Having uploader and viewer be both separate permissions helps to enforce the least privilege principle. If one of the roles get compromised, there is little risk of having both the method and content of uploaded materials be exploited. In other words, by splitting the read and write permissions to two different roles, a hacker can either overwrite shared.txt without reading its contents or read the contents of the uploaded file without the ability to modify said file. Splitting roles reduces the maximum impact of one of the roles being compromised.
 13. One or two sentences explaining why embedding code directly into user-data means the instance never needs a Git or S3 credential just to fetch its own code.
 - a. Using Git or S3 would imply that the instance launches without application code; by launching with embedded app code, the EC2 instance can begin operation immediately using just launch template user data.